

COMS30035, Machine learning: Regression and Classification Trees

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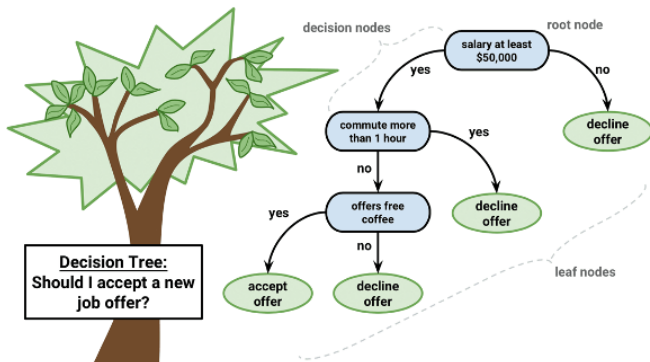
Acknowledgement

- ▶ These slides are adapted from ones originally created by Edwin Simpson.

Decision Trees

A Decision Tree is a tree-structured predictive model where:

- ▶ Each internal node represents a decision test on a feature (e.g., Is salary $> 50k$?).
- ▶ Each branch represents the outcome of the test (e.g., yes/no).
- ▶ Each leaf node represents a predicted class label (for classification) or a numerical value (for regression).



Decision Trees as Partitioning Input Space

- ▶ One model is responsible for assigning a decision for each region of input space;
- ▶ The correct model for an input $\mathbf{x}$ is chosen by traversing the binary decision tree, following the path from the top to a leaf.
- ▶ Leaf node is responsible for assigning a decision, such as a:
 - ▶ Class label;
 - ▶ Probability distribution over class labels;
 - ▶ Scalar value (for regression tasks).

Learning the Tree Structure

- ▶ Which input variable to use at each node?
- ▶ How to determine the splitting value for the chosen variable?
- ▶ Classification and Regression Trees (CART): one of many possible learning algorithms
- ▶ Objective: greedily minimise the error
 - ▶ Regression: sum-of-squares
 - ▶ Classification: cross-entropy as used in neural networks or Gini impurity

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Learning the Tree Structure

- ▶ Number of possible solutions grows combinatorially with the number of input variables
- ▶ Greedy algorithm: add nodes one-at-a-time, choosing the best split at each point
 1. Start from the root node
 2. Run *exhaustive search* over each possible variable and each possible threshold (splitting value) for that variable. For each variable and threshold:
 - ▶ Compute average of the target variable for each leaf of the proposed node
 - ▶ Compute the error if we stop adding nodes here
 3. Choose the variable & threshold that minimise the error
 4. Add a new node for the chosen variable and threshold.
 5. Repeat step 2 until there are only n data points associated with each leaf node.
 6. Prune back the tree to remove branches that do not reduce error by more than a small tolerance value, ϵ .

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Splits at a node in the tree

We will be following the notation and explanation given by Murphy since this is clearer than Bishop's (although there are typos).

- ▶ Suppose we are considering the best split for node i in a tree.
- ▶ Let $\mathcal{D}_i$ be the datapoints which reach node i .
- ▶ If the j th feature is a real-value scalar then a split on threshold t looks like:

$$\begin{aligned}\mathcal{D}_i^L(j, t) &= \{(\mathbf{x}_n, y_n) \in \mathcal{D}_i : x_{n,j} \leq t\} \\ \mathcal{D}_i^R(j, t) &= \{(\mathbf{x}_n, y_n) \in \mathcal{D}_i : x_{n,j} > t\}\end{aligned}$$

- ▶ We need only consider thresholds t that are values of feature j observed in the data.
- ▶ If the j th feature is categorical, a split on value t looks like:

$$\begin{aligned}\mathcal{D}_i^L(j, t) &= \{(\mathbf{x}_n, y_n) \in \mathcal{D}_i : x_{n,j} = t\} \\ \mathcal{D}_i^R(j, t) &= \{(\mathbf{x}_n, y_n) \in \mathcal{D}_i : x_{n,j} \neq t\}\end{aligned}$$

- ▶ NB. Murphy sometimes has N_i when it should be $\mathcal{D}_i$.

The best possible split

The best possible split is:

$$(j_i, t_i) = \arg \min_{j \in \{1, \dots, D\}} \min_{t \in \mathcal{T}_j} \frac{|\mathcal{D}_i^L(j, t)|}{|\mathcal{D}_i(j, t)|} c(\mathcal{D}_i^L(j, t)) + \frac{|\mathcal{D}_i^R(j, t)|}{|\mathcal{D}_i(j, t)|} c(\mathcal{D}_i^R(j, t))$$

where i is the current node, D is the number of features, $\mathcal{T}_j$ is the set of possible thresholds for feature j and c is the cost function.

For regression, we typically choose c to be mean squared error:

$$c(\mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{n \in \mathcal{D}} (y_n - \bar{y})^2$$

where $\mathcal{D}$ is the set of datapoints reaching the leaf and $\bar{y} = \frac{1}{|\mathcal{D}|} \sum_{n \in \mathcal{D}} y_n$ is the mean of the response variable for datapoints in $\mathcal{D}$. (NB: there is a typo in Murphy for that last equation).

Cost functions for classification

- First compute the *empirical distribution* for the data in a leaf:

$$\hat{\pi}_c = \frac{1}{|\mathcal{D}|} \sum_{n \in \mathcal{D}} \mathbb{1}(y_n = c)$$

One possible cost function is the *Gini index*:

$$G = \sum_{i=1}^C \hat{\pi}_i (1 - \hat{\pi}_i) = 1 - \sum_{i=1}^C \hat{\pi}_i^2$$

Another is *entropy*

$$H = - \sum_{i=1}^C \hat{\pi}_i \log \hat{\pi}_i$$

Example of Information Gain Calculation

We can view using entropy as trying to maximise the *information gained* by choosing a split.

A	B	C	Class
yes	yes	yes	3
yes	yes	no	2
yes	no	yes	3
yes	no	no	1
no	yes	yes	3
no	yes	no	2
no	no	yes	3
no	no	no	1

Step 1: calculate start entropy:

$$E_{start}(S) = -\frac{2}{8}\log_2\left(\frac{2}{8}\right) - \frac{2}{8}\log_2\left(\frac{2}{8}\right) - \frac{4}{8}\log_2\left(\frac{4}{8}\right) = 0.5 + 0.5 + 0.5 = 1.5 \text{ bits}$$

Step 2: entropy of the training data if S were divided into S1 and S2: i.e. for attribute A:

$$E_A(S) = \frac{4}{8}\left(-\frac{1}{4}\log_2\left(\frac{1}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)\right) + \frac{4}{8}\left(-\frac{1}{4}\log_2\left(\frac{1}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)\right) = 1.5 \text{ bits}$$

Repeat for all attributes

Step 3: calculate information gain for all possible divisions: i.e. for attribute A:

$$Gain(A) = E(S) - E_A(S) = 1.5 \text{ bits} - 1.5 \text{ bits} = 0 \text{ bits}$$

Example of Information Gain Calculation

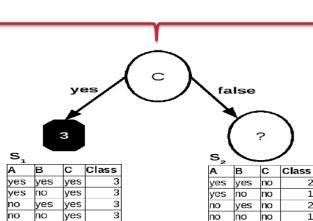
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The same calculation is performed also for the remaining attributes resulting in information gains of:

$$Gain(A) = E(S) - E_A(S) = 1.5 \text{ bits} - 1.5 \text{ bits} = 0 \text{ bits}$$

$$Gain(B) = E(S) - E_B(S) = 1.5 \text{ bits} - 1 \text{ bit} = 0.5 \text{ bits}$$

$$Gain(C) = E(S) - E_C(S) = 1.5 \text{ bits} - 0.5 \text{ bits} = 1 \text{ bit}$$



Stopping Criteria in Decision Trees

- ▶ Growing a tree without limits often leads to **overfitting**.
- ▶ We need **stopping criteria** to decide when to stop splitting:
 - ▶ Maximum tree depth reached;
 - ▶ Minimum number of samples in a leaf node;
 - ▶ Minimum number of samples required to split a node;
 - ▶ Impurity (e.g. Gini, entropy, MSE) below a threshold;
 - ▶ No further information gain from any split.
- ▶ These rules help balance **model complexity** and **generalization**.

Pruning

- ▶ Balance residual training-set error against model complexity
- ▶ Start with a tree T_0
- ▶ Consider pruning each node in T_0 by combining the branches to obtain tree T
- ▶ Compute a criterion $C(T) = \sum_{\tau=1}^{|T|} e_{\tau}(T) + \lambda|T|$
- ▶ If $C(T) \leq C(T_0)$ keep the pruned tree, else reinstate the pruned node.
- ▶ The hyperparameter λ can be chosen using a validation set or via cross-validation

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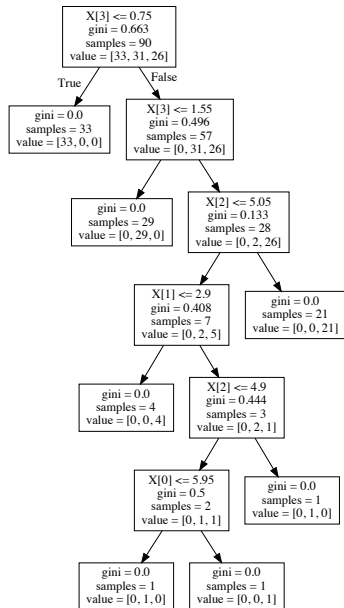
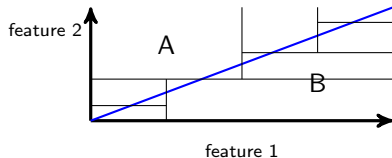
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Interpretability

- ▶ The sequence of decisions is often easier to interpret than other methods (think of neural networks);
- ▶ However, sometimes small changes to the dataset cause big changes to the tree;
- ▶ If the optimal decision boundary is not aligned with the axes of an input variable, we need a lot of splits.



Pros and Cons of CART

Pros

- ▶ Easy to interpret and visualize
- ▶ Handles both classification and regression
- ▶ Non-parametric (no strong assumptions)
- ▶ Works with numerical and categorical features
- ▶ Captures nonlinear relationships
- ▶ Robust to scaling of features

Cons

- ▶ Prone to overfitting (needs pruning)
- ▶ High variance (unstable to small data changes)
- ▶ Only axis-aligned splits
- ▶ Biased toward features with many split points
- ▶ Single tree is often less accurate than ensembles
- ▶ Regression outputs are piecewise constant

Ensemble Methods

- ▶ **Bagging (Bootstrap Aggregating)**

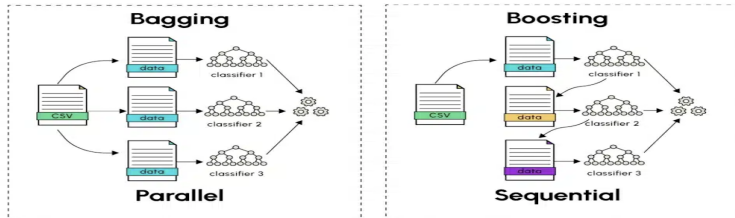
Train multiple trees on bootstrapped samples; average or vote.
(*Reduces variance*).

- ▶ **Random Forests**

Extension of Bagging with random feature subsets at splits. (*Stable, accurate*).

- ▶ **Boosting (AdaBoost, Gradient Boosting, XGBoost)**

Sequentially build trees, focusing on correcting errors (reduces bias).
(*High accuracy*).



<https://datascientest.com/en/bagging-vs-boosting>

Reading

There is typo on p. 666 of Bishop where a minus sign has gone missing.
Equation (14.32) should be:

$$Q_{\tau}(T) = - \sum_{k=1}^K p_{\tau k} \ln p_{\tau k}$$

- ▶ Bishop §14.4.
- ▶ Murphy §18.1

Problems and quizzes

- ▶ Bishop Exercise 14.10
- ▶ Bishop Exercise 14.11 (don't forget about the typo!)
- ▶ Quizzes:
 - ▶ Week 3: Trees